

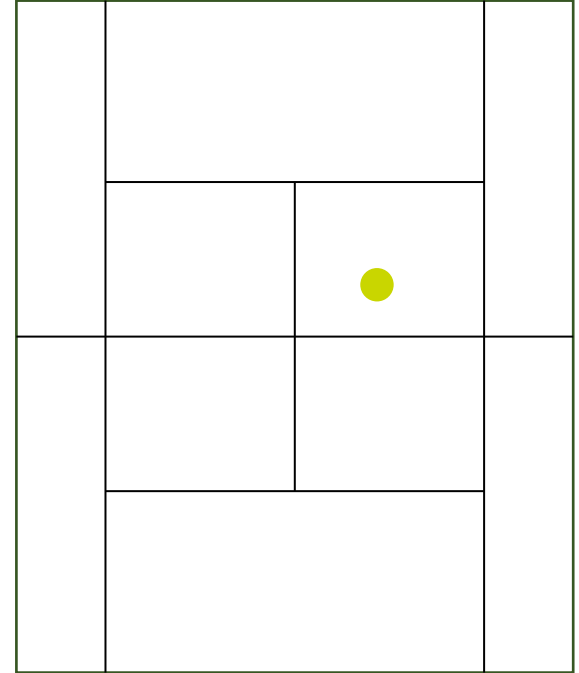
GROUP D

Fatigue, Form, and the Efficiency of Tennis Rankings

Predicting ATP match outcomes (1991–2024) as an information-efficiency test

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Data & AI in Economics · TU Dortmund





Motivation & Economic Relevance

The ranking is a market price. The ATP ranking aggregates every player's recent results into the sport's consensus signal of quality, the analogue of a market price that impounds public information.

So beating it is an efficiency test. If a model using only pre-match information can beat “higher-ranked player wins,” the ranking does **not** fully impound public information, a weak-form **Efficient-Market-Hypothesis** test, run in a clean setting where every outcome is settled and unambiguous.

WHY IT MATTERS ECONOMICALLY

\$147 bn

wagered on sport in the
U.S. in 2024 (AGA)

#1

fastest-growing betting
sport: tennis

Odds are priced off rankings & form. Any non-priced signal (fatigue, style) is, in principle, an exploitable edge and a test of how efficiently public information is aggregated.

Beating the ranking benchmark tests market efficiency in a clean, settled-outcome laboratory — with direct pricing relevance for a \$100 bn+ betting market.



Data & Methodology

108k

tour-level matches

34

seasons (1991–2024)

≤ 2019

train · 92,110

≥ 2022

test · 13,158



Causal

Does fatigue lower the next-match win probability?

DAG + propensity-score weighting
(DoWhy, backdoor) + 3 refutation tests.



Supervised

Can a model beat the ranking benchmark?

Logistic regression vs. random forest vs. baseline — chronological CV; accuracy, log-loss, AUC, McNemar.



Unsupervised

Are there playing-style archetypes?

K-Means + Ward clustering on serve/return profiles → a leakage-safe predictive feature.



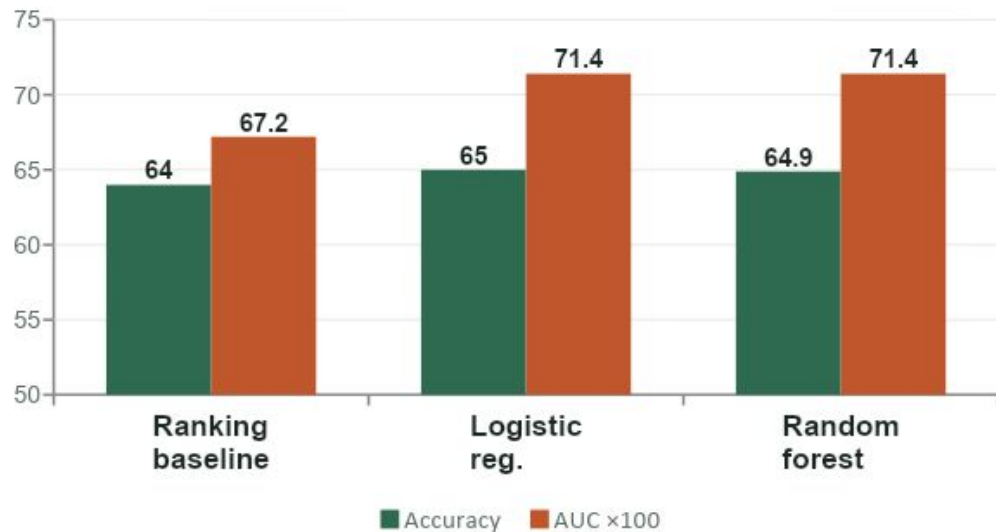
Leakage-safe by design: chronological split with 2020–21 held out as a COVID washout · strictly backward-looking features · clustering fit on training years only · treatment threshold learned on training data.



RESULTS

Can a model beat the ranking?

Test set 2022–2024 — accuracy & AUC (×100)



+1.0 pp

model accuracy over
baseline

0.714

AUC vs. 0.672 baseline

$p < .01$

McNemar: significant ($p = .0016$)

65.7%

chronological CV (± 1.5)

50 = coin-flip (no skill)

TAKEAWAY The ranking is near-efficient but NOT fully efficient: a model beats it by a small, statistically significant margin (AUC 0.67 → 0.71). A weak-form inefficiency — real, but modest.

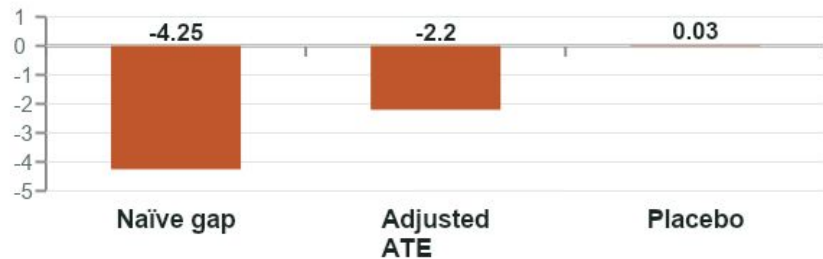


RESULTS

Why? Fatigue and playing style



Fatigue · causal



-2.2 pp lower win prob. after a long previous match.
95% CI [-2.84, -1.63] · stable under refuters · placebo $\approx$ 0.



Playing style · archetypes

Players split along a **server** $\leftrightarrow$ **returner** axis (4 archetypes, by ace rate):

13.3%

Big server

7.0%

Balanced

6.7%

All-court

3.6%

Returner

Used as a previous-season (leakage-safe) feature. K-Means vs. Ward agree (ARI 0.29); the archetype ranks among the model's predictive features.

TAKEAWAY Both fatigue (a real causal -2.2 pp effect) and playing style carry information the ranking alone misses — the mechanisms behind the inefficiency.



Limitations & Synthesis

SYNTHESIS — THE EFFICIENCY VERDICT

Near-efficient, not perfectly efficient.

The ranking captures most public information (64% baseline), yet logistic & RF models beat it by ~1 pp (AUC 0.67 → 0.71, $p < .01$).

The edge is mechanistic.

It traces to fatigue (causal -2.2 pp) and, marginally, playing style — signals not fully priced into the ranking.

Economic parallel.

As in financial markets, the public signal is mostly but not perfectly efficient: small, identifiable, exploitable deviations persist.



LIMITATIONS & THREATS TO VALIDITY

- **Small effect sizes**
 - the inefficiency is real but modest; bookmaker margins likely erase exploitability.
- **Pre-match information only**
 - no in-match dynamics, injuries, or surface-specific motivation.
- **Ranking $\neq$ all public info**
 - betting odds would be a sharper efficiency benchmark.
- **Noisy archetypes**
 - silhouette favours $k = 2$ ($k = 4$ used for interpretability); ARI 0.29 = moderate method agreement.
- **Causal identification**
 - rests on a no-unobserved-confounding (backdoor) assumption.

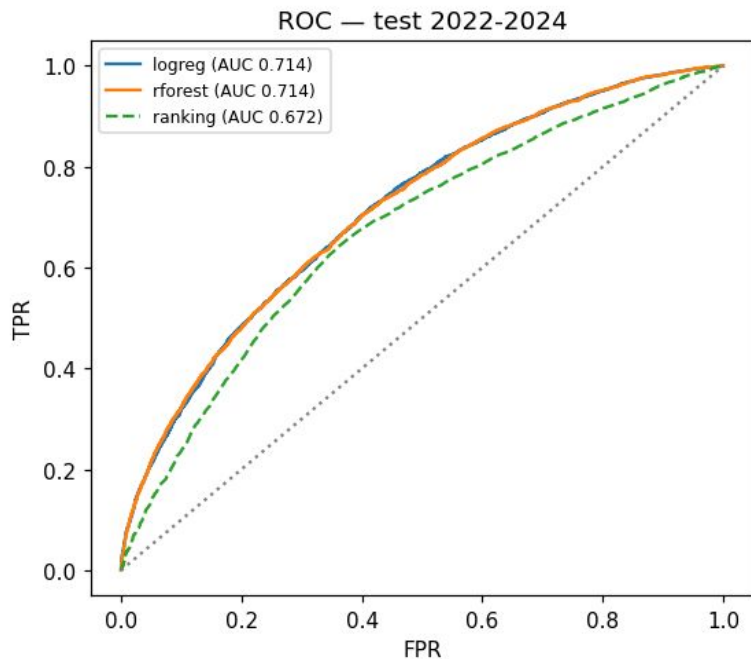


References

1. Pedregosa et al. (2011). Scikit-learn: ML in Python. JMLR 12, 2825–2830.
2. Breiman, L. (2001). Random forests. Machine Learning 45(1), 5–32.
3. Hastie, Tibshirani & Friedman (2009). The Elements of Statistical Learning (2nd ed.). Springer.
4. Rosenbaum & Rubin (1983). The central role of the propensity score. Biometrika 70(1), 41–55.
5. Pearl, J. (2009). Causality (2nd ed.). Cambridge University Press.
6. Sharma & Kiciman (2020). DoWhy: an end-to-end library for causal inference. arXiv:2011.04216.
7. Ward, J. H. (1963). Hierarchical grouping to optimize an objective function. JASA 58(301), 236–244.
8. MacQueen, J. (1967). Methods for classification of multivariate observations. Berkeley Symp. 1, 281–297.
9. Klaassen & Magnus (2001). Are points in tennis i.i.d.? JASA 96(454), 500–509.
10. Klaassen & Magnus (2003). Forecasting the winner of a tennis match. EJOR 148(2), 257–267.
11. Boulier & Stekler (1999). Are sports seedings good predictors? Int. J. Forecasting 15(1), 83–91.

Data: Jeff Sackmann, *tennis_atp* — yearly atp_matches_YYYY.csv, tour-level from 1991. © Jeff Sackmann / Tennis Abstract, CC BY-NC-SA 4.0; vendored in the repo under data/ (original repository made private in 2025). Sackmann's original public repository was made private in 2025; the atp_matches_YYYY.csv files used here are therefore vendored in this repository under data/ to keep the analysis self-contained and reproducible <https://github.com/Frankz04/dai-mission-group-D/tree/main/data>.

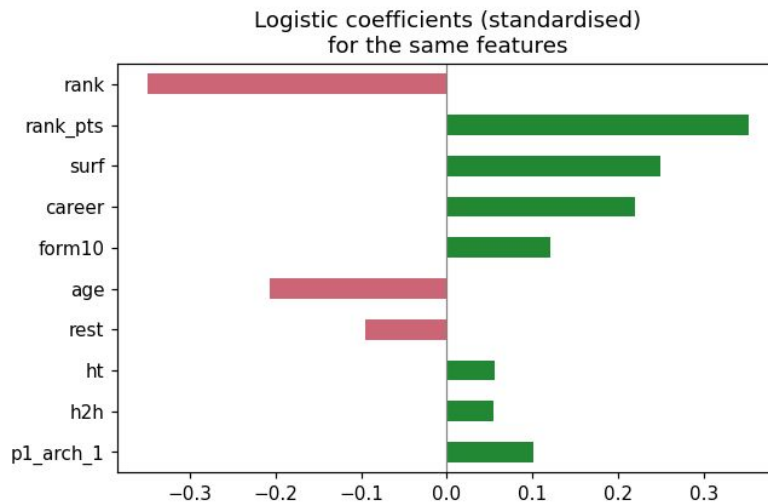
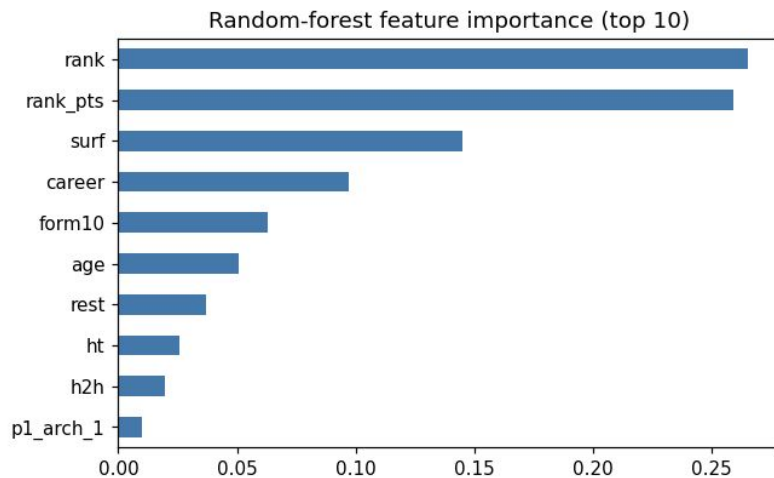
Discrimination: ROC and test-set metrics



Model	Acc.	Log-loss	AUC
Ranking baseline	0.640	—	0.672
Logistic reg.	0.650	0.619	0.714
Random forest	0.649	0.618	0.714

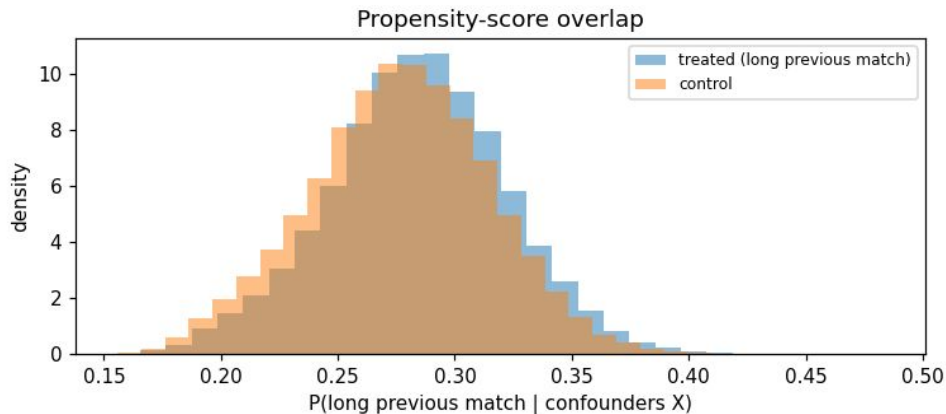
McNemar (logistic vs. baseline): $p = 0.0016$ · chronological CV 65.7% $\pm 1.5\%$ · test 2022–2024 ($n = 13,158$).

What drives prediction: importances & coefficients



Rank and rank-points dominate; recent form, surface, career win-rate and the playing-style archetype (p1_arch) supply the marginal edge over the ranking. Signs match the causal story (longer rest $\uparrow$, older age $\downarrow$).

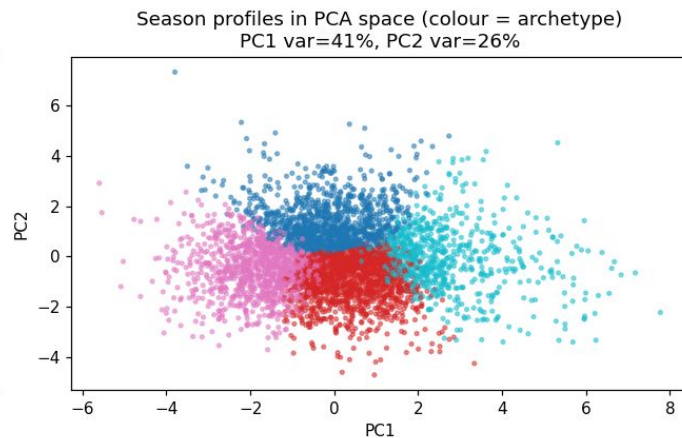
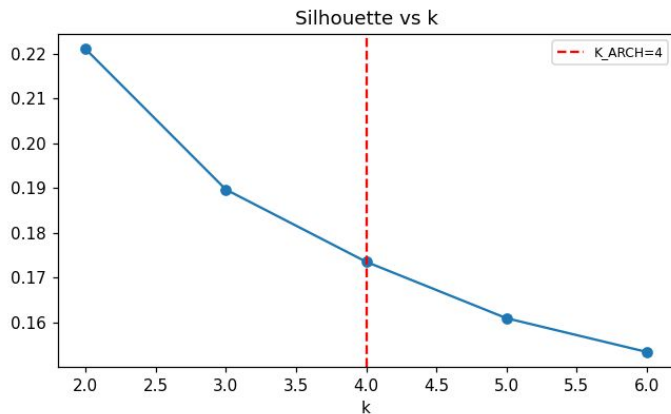
Fatigue effect: overlap and ATE estimates



Estimator	ATE (pp)
Naïve (unadjusted)	-4.25
Manual IPW	-2.20
Stratification	-2.36
DoWhy IPW	-2.00
Placebo (permuted)	+0.03

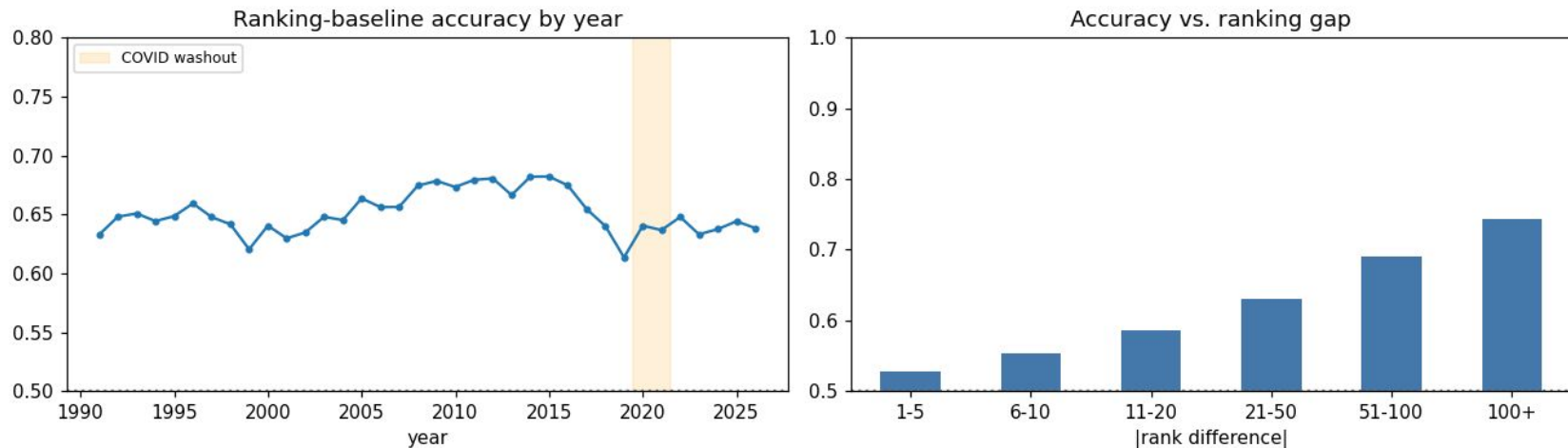
Manual IPW 95% CI $[-2.84, -1.63]$ pp · stable under random-common-cause and 80%-subset refuters · $n = 113,798$, treated 27.9%.

Playing-style archetypes (serve/return profiles)



Archetype	n	Ace rate	1st-serve in	1st-won	Return-won
Big server / shotmaker	692	13.3%	59.4%	77.3%	34.3%
Balanced baseliner	1537	7.0%	60.2%	71.8%	37.9%
All-court	1115	6.7%	56.4%	70.8%	37.0%
Returner / counterpuncher	1055	3.6%	63.9%	65.6%	38.9%

Ranking accuracy is stable; predictability rises with the rank gap



Left: the ~64–68% ranking baseline is stable across 1991–2024 (2020–21 washout shaded). Right: matches between closely-ranked players are near coin-flips (~53%), while large rank gaps are highly predictable (~74%).